MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**
Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3670 PRIMARY CYANIDE DESTRUCTION
DESIGN OF TANK FOUNDATION

Document Number: **MD-216-3670-EG-CV-CAL-1355**
Revision: **D01**
Revision Date: **27.11.2023**
Status: **AS BUILT**

Outotec Document Number:	900707	GAU01	RN12	CED01	36702	OU100672878	2
L&T Document Number:	O19045	450	C	K	24144	001	2

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	12.03.2020	FIRST ISSUE	SOR	SJM	AG	
C01	06.05.2020	ISSUED FOR CONSTRUCTION	SOR	SJM	AG	
D01	27.11.2023	AS BUILT	SOR	SJM	AG	

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DEFINITIONS AND ABBREVIATIONS

Document	:	Information and the medium on which it is contained (ISO 9000:2015)
EMPLOYER	:	'EMPLOYER' means MA'ADEN GOLD AND BASE METALS (MGBM) and the PMC American Bechtel Inc. or their representatives
Contractor	:	Contractor' means consortium composed of LARSEN & TOUBRO LIMITED, Saudi Arabia AND OUTOTEC SAUDI ARABIA LLC, Saudi Arabia (for IK Contract) and LARSEN & TOUBRO LIMITED – ECC Construction Group – (SHJ BR) and OUTOTEC ENGINEERING DMCC, Dubai (for OOK Contract).
'PMT'	:	Ma'aden Project Management Team
'MMGP Project'	:	Mansourah Massarah Gold Processing Project
Attribute	:	Information element that describes a document or tag.
Supplier	:	An entity (supplier, consultant, sub-contractor) under the direct project management of Contractor
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1. INTENT

This document is to Prepare for design of **Primary Cyanide Destruction (3670)-** Tank Foundation. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2.EXECUTIVE SUMMARY

The Saudi Arabian Mining Company (Ma'aden) has announced that its wholly-owned subsidiary, Ma'aden Gold and Base Metals Company, have issued a Notice of Award to Outotec and Larsen & Toubro Consortium to complete the engineering, procurement, construction, pre-commissioning, commissioning and start-up assistance of the Mansourah & Massarah (MMGP) Gold Project.

MMGP project will become Ma'aden's largest ever gold project, producing an average of 250,000 ounces of gold per year over the life of mine. Mansourah-Massarah is situated within the Central Arabian Gold Region (CAGR), approximately 320km northeast of Taif towards Riyadh and about 35 km southeast of Hufairah, which is a small village situated on Jeddah - Riyadh highway. Mansourah - Massarah gold deposits are located 22°38'6.40"N; 42°44'22.90"E, and 22°42'1.40"N; 42°41'58.40"E respectively, approximately 537 km east of Jeddah and 77 km southeast of the town of Zalim.

The plant throughput capacity is 4000000t/a and is required to operate on a continuous basis at nominally 24 hours per day and 7 days a week.

3. SCOPE

The analysis has been carried out in Excel and Staad. For member strength calculation force values are directly taken from Load Data Provided & Output of Staad analysis.

4. DESIGN REQUIREMENTS

4.1 Design Loads

All Design Loads are indicated in Foundation and Load Plan (Doc. No. MD-216-3670-EG-ME-PLN- 1001) attached in Attachment 1. Also indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

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5. CONCRETE FOUNDATIONS

5.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

5.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 Mpa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric : ASTM A 185(Plain) / A 497(Deformed)

5.3 Foundations for Aboveground Tanks

Foundation shall be Raft Foundation with a Concrete Ringwall.

5.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50
	Raft	50

5.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered as 450 kN/m^2 at 1.2m depth below EGL (ie. RL 911.00m) refer Table 5 - BH-27 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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6. DESIGN CALCULATION FOR FOUNDATION

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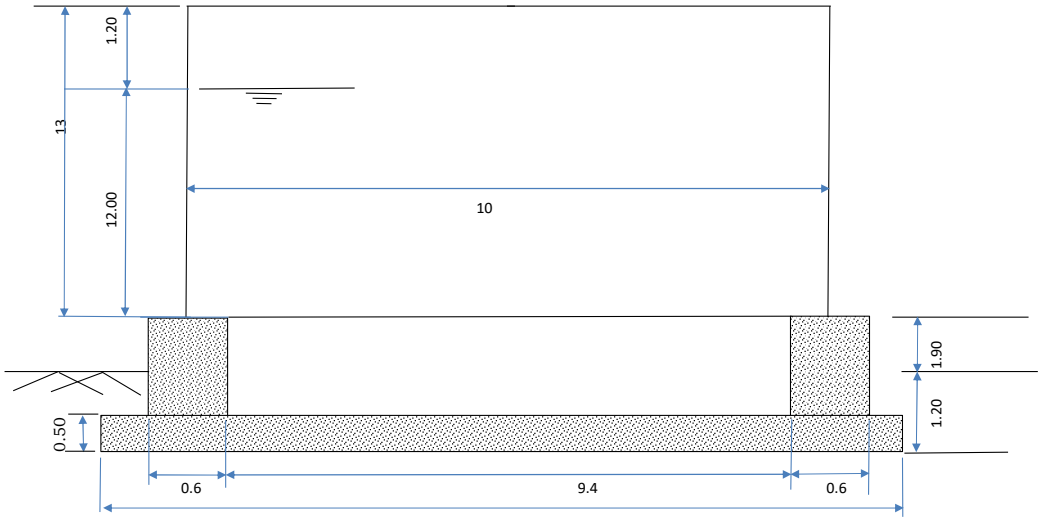
Foundation Marked F1 , F2 & F3

Tank Loading Data

DEAD WEIGHT OF EQUIPMENT	640	kN
DEAD WEIGHT OF CIVIL STRUCTURES	1250	kN
OPERATIONAL WEIGHT OF SLURRY	13161	kN
MAXIMUM WEIGHT OF SLURRY	14491	kN
EXTERNAL PLATFORM LOAD	100	kN
WIND LOAD	154.7	kN
SEISMIC LOAD	643	kN
SEISMIC TORQUE AT FOUNDATION LEVEL	5988	kNm

	Loading Type		
Weight, V	Empyt (W)	1990	kN
	Operating (Wo)	15151	kN
	Max (Wt)	16481	kN

Summary of Tank Specification



Diameter of Tank, D	=	10 m	
Height of Tank, H	=	13.2 m	
Height of Tank Fluid, H_L	=	12 m	
Inner Diameter of Concrete Ring Wall, ID	=	9.4 m	
Outer Diameter of Concrete Ring Wall, OD	=	10.6 m	
Height of Concrete Ring Wall, h	=	3.10 m	
Thickness of Concrete Ring Wall, t	=	0.6 m	
Thickness of Concrete Raft, T	=	0.5 m	
P.C.D of Bolts	=	10.272 m	
Top of Foundation	=	1.9 m	(Above Finished Ground Level)
Unit Weight of Liquid, γ_L	=	15.4 kN/m^3	
Unit Weight of Soil, γ_s	=	20 kN/m^3	
Unit Weight of Concrete, γ_c	=	24 kN/m^3	

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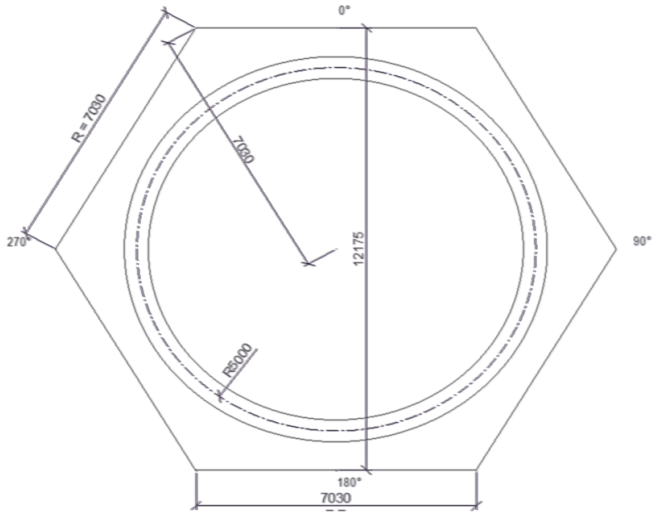
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Wind Load & Seismic Load

Total Base Shear due to wind load = 154.7 kN
Total Base Shear due to Seismic load = 643 kN

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	154.70	1191.19
	Operating	154.70	1191.19
Seismic Loads	Operating	643.00	5988.00



Foundation Weight

a) Ring Wall Weight = 1176.21 kN
b) Inside Soil Weight = 3608.68 kN
c) Outside Soil Weight = 562.07 kN
d) Raft Weight = 1540.75 kN

Area of Raft

Area of Raft (Hexagonal), $A = 2.598 * r^2$ = 128.40 m^2

Property of Hexagon
Radious of outer circle, r = 7.03 m
Dist. Of 2 opposite side, d = 12.175 m

Sectional Modulus of Raft

Sectional modulus (Hexagonal), $Ze = 0.0601*d^4 / r$ = 187.84 m^3

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 , Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	15151.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	1194.00	154.70	1191.19
	Operating	LC103	D + F1 + W	15151.00	154.70	1191.19
		LC104	D + F1+ 0.7 E	15151.00	450.10	4191.60
		LC105	0.6 (D + F1) + 0.7 E	9090.60	450.10	4191.60
	Maximum weight	LC106	D + F2	16481.00	0.00	0.00

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Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 450 KN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 598.5 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	22038.72	0.00	171.65	171.65	2.62	OK
LC102	8081.72	1670.76	71.84	54.05	8.33	OK
LC103	22038.72	1670.76	180.54	162.75	3.32	OK
LC104	22038.72	5586.91	201.39	141.90	2.97	OK
LC105	15978.32	5586.91	154.19	94.70	3.88	OK
LC106	23368.72	0.00	182.01	182.01	2.47	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
	V (kN)					
LC101	22038.72	0.3	6611.62	0.00	-	-
LC102	8081.72	0.3	2424.52	154.70	15.7	OK
LC103	22038.72	0.3	6611.62	154.70	42.7	OK
LC104	22038.72	0.3	6611.62	450.10	14.7	OK
LC105	15978.32	0.3	4793.50	450.10	10.6	OK
LC106	23368.72	0.3	7010.62	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of	Edge Dis.(d/2)	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
	V (kN)					
LC101	22038.72	6.0875	134160.69	0.00	-	-
LC102	8081.72	6.0875	49197.45	1670.76	29.4	OK
LC103	22038.72	6.0875	134160.69	1670.76	80.3	OK
LC104	22038.72	6.0875	134160.69	5586.91	24.0	OK
LC105	15978.32	6.0875	97268.01	5586.91	17.4	OK
LC106	23368.72	6.0875	142257.07	0.00	-	-

Ultimate Strength Design

Load Combination (As per SBC-304 , Table 5.3.1)

Term of Load	Load Conditions	Load Case	Load Combination	Load at bottom of tank Inner Area			
				V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC201	1.4 (D + F1)	21211.40	0.00	78.54	98.17
Short term	Empty	LC202	0.9 D + 1.6 W	1791.00	1905.90	78.54	98.17
	Operating	LC203	1.2 (D + F1) + 1.6W	18181.20	1905.90	78.54	98.17
		LC204	1.2 (D + F1) + E	18181.20	5988.00	78.54	98.17
		LC205	0.9 (D + F1) + E	13635.90	5988.00	78.54	98.17
	Test	LC206	1.4 (D + F2)	23073.40	0.00	78.54	98.17

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Staad input calculation.

Total Number of Noads in Wall Top = 52

S/W of tank (Empty Condition) on each Nodes= 38.27 kN

Pressure due to Sand fill on inside Raft= 52.0 kN/m^2

Pressure due to soil fill on outside Raft= 14.0 kN/m^3

Wind Base Shear of tank on each Nodes= 2.975 kN

Seismic Base Shear of tank on each Nodes= 12.365 kN

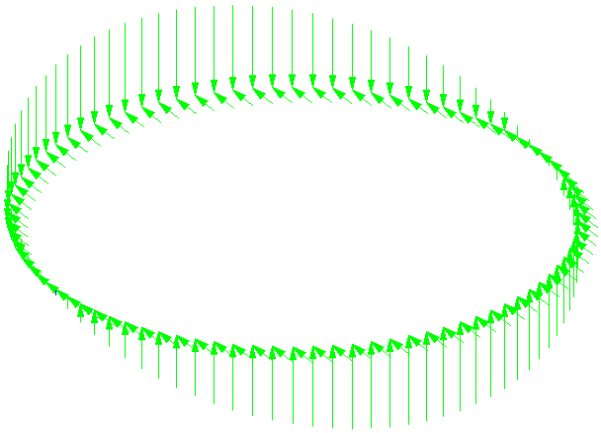
Sl No.	Node No.	Dist. R(m)	R^2	Seismic V(kN)	Wind V(kN)
1	2487	0.00	0.00	0.0	0.0
2	2488	-0.60	0.36	-5.6	-1.1
3	2489	-1.20	1.43	-11.0	-2.2
4	2490	-1.77	3.14	-16.3	-3.2
5	2491	-2.32	5.40	-21.4	-4.3
6	2492	-2.84	8.07	-26.2	-5.2
7	2493	-3.32	10.99	-30.5	-6.1
8	2494	-3.74	14.01	-34.5	-6.9
9	2495	-4.11	16.93	-37.9	-7.5
10	2496	-4.43	19.60	-40.8	-8.1
11	2497	-4.68	21.86	-43.1	-8.6
12	2498	-4.85	23.57	-44.7	-8.9
13	2499	-4.96	24.64	-45.7	-9.1
14	2500	-5.00	25.00	-46.1	-9.2
15	2501	-4.96	24.64	-45.7	-9.1
16	2502	-4.85	23.57	-44.7	-8.9
17	2503	-4.68	21.86	-43.1	-8.6
18	2504	-4.43	19.60	-40.8	-8.1
19	2505	-4.11	16.93	-37.9	-7.5
20	2506	-3.74	14.01	-34.5	-6.9
21	2507	-3.32	10.99	-30.5	-6.1
22	2508	-2.84	8.07	-26.2	-5.2
23	2509	-2.32	5.40	-21.4	-4.3
24	2510	-1.77	3.14	-16.3	-3.2
25	2511	-1.20	1.43	-11.0	-2.2
26	2512	-0.60	0.36	-5.6	-1.1
27	2513	0.00	0.00	0.0	0.0
28	2462	0.60	0.36	5.6	1.1
29	2463	1.20	1.43	11.0	2.2
30	2464	1.77	3.14	16.3	3.2
31	2465	2.32	5.40	21.4	4.3
32	2466	2.84	8.07	26.2	5.2
33	2467	3.32	10.99	30.5	6.1
34	2468	3.74	14.01	34.5	6.9
35	2469	4.11	16.93	37.9	7.5
36	2470	4.43	19.60	40.8	8.1
37	2471	4.68	21.86	43.1	8.6
38	2472	4.85	23.57	44.7	8.9
39	2473	4.96	24.64	45.7	9.1
40	2474	5.00	25.00	46.1	9.2
41	2475	4.96	24.64	45.7	9.1
42	2476	4.85	23.57	44.7	8.9
43	2477	4.68	21.86	43.1	8.6
44	2478	4.43	19.60	40.8	8.1
45	2479	4.11	16.93	37.9	7.5
46	2480	3.74	14.01	34.5	6.9
47	2481	3.32	10.99	30.5	6.1
48	2482	2.84	8.07	26.2	5.2
49	2483	2.32	5.40	21.4	4.3
50	2484	1.77	3.14	16.3	3.2
51	2485	1.20	1.43	11.0	2.2
52	2486	0.60	0.36	5.6	1.1
		ΣR^2=	**650.00**		

Staad input calculation For Seismic and Wind load distribution.

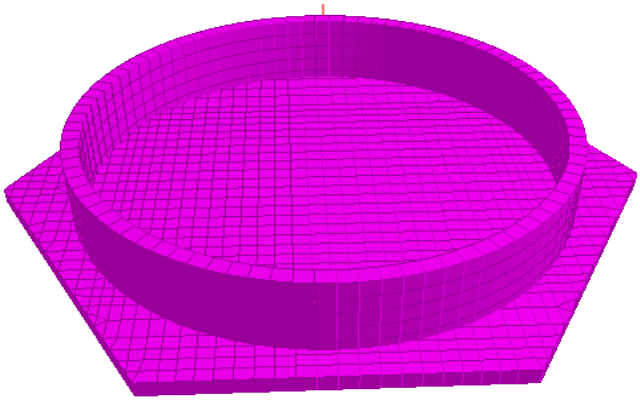
$V_{wind} = R * W_{total} / \Sigma R^2$ $\quad H_{wind} = W_{total} / 90$

$V_{Seismic} = R * E_{total} / \Sigma R^2$ $\quad H_{seismic} = E_{total} / 90$

H_{wind}=	2.975	kN
$H_{seismic}$=	12.365	kN



Staad Model in 3D view



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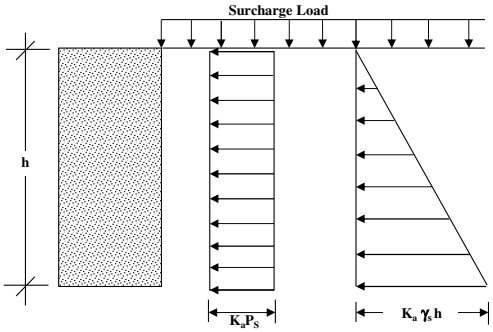
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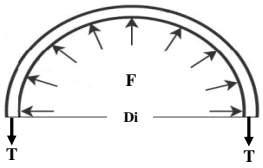
Wall Design

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m²
LC201	21211.40	0.00	270.07
LC202	1791.00	1905.90	42.22
LC203	18181.20	1905.90	250.90
LC204	18181.20	5988.00	292.48
LC205	13635.90	5988.00	234.61
LC206	23073.40	0.00	293.78



Maximum Surcharge Load, Ps = 293.78 kN/m²



Where, K_a = coefficient of earth pressure at rest = 0.5

Pressure due to Surcharge,P1 = $K_a * P_s$ = 146.89 KN/m²

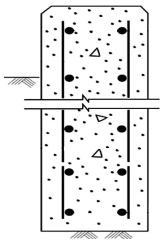
Pressure due to Soil,P2 = $1.4 * \gamma_S * (h-T) * K_a$ = 36.40 KN/m² for LC206 - 1.4 (D + F2)

Maximum Lateral Load

F = P1 * (h-T) + 1/2 * P2 *(h -T)= 429.2335 KN/m

Hoop Tension Load

T= 1/2 * F * ID = 2017.398 KN



Hoop Tension Bar

1. Horizontal Reinfocement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(ϕ x Fy) = 5337.031 mm²

Minimum As = 0.0025 x t x h = 3900 mm² (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 32 Nos. D20= 9072.92 mm² OK

2. Vertical Reinfocement

Minimum As = 0.0015 * b * D = 900 mm² (As per API 650 , Cl. No. B.4.2.3)

STAAD output:- Wall Thickness = 600 mm

Characteristic strength of concrete, f'_c = 35 MPa

Yield stress of steel reinforcement, f_y = 420 MPa

Clear cover for main reinforcement bars = 50mm

Max BM (+) (M_y)=	141.792	kN m/m	(VERTICAL MOMENT IN INSIDE FACE)
Max BM (-) (M_y)=	8.326	kN m/m	(VERTICAL MOMENT IN OUTSIDE FACE)
Max BM (+) (M_x)=	28.865	kN m/m	(HORIZONTAL MOMENT IN INSIDE FACE)
Max BM (-) (M_x)=	4.381	kN m/m	(HORIZONTAL MOMENT IN OUTSIDE FACE)

Max Shear Qx = 0.026 N/mm²

Max Shear Qy = 0.371 N/mm²

Effective depth of wall in Vertical direction = 542 mm

Effective depth of wall in Horizontal direction = 524 mm

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Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	141.79	0.48	0.64	0.002	839.69	975.60	D16	206.09	300	175	1148.93
My(-)	8.33	0.03	0.04	0.000	48.80	975.60	D16	206.09	300	175	1148.93
Mx(+)	28.87	0.11	0.14	0.000	175.29	975.60	D20	322.02	300	175	1795.20
Mx(-)	4.38	0.02	0.02	0.000	26.55	975.60	D20	322.02	300	175	1795.20

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.37 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN WALL: **600 mm**

VERTICAL REIN. IN INSIDE FACE	**D16**	**175**	**c/c**	
VERTICAL REIN. IN OUTSIDE FACE	**D16**	**175**	**c/c**	
HORIZONTAL REIN. IN INSIDE FACE	**D20**	**175**	**c/c**	**(16Nos.)**
HORIZONTAL REIN. IN OUTSIDE FACE)	**D20**	**175**	**c/c**	**(16Nos.)**

Raft Design

STAAD output:- Raft Thickness = 500 mm

Max BM (+) (M_y)=	117.649	kN m/m (HORIZONTAL MOMENT IN TOP FACE)
Max BM (-) (M_y)=	132.597	kN m/m (HORIZONTAL MOMENT IN BOTTOM FACE)
Max BM (+) (M_x)=	99.363	kN m/m (LONGITUDINAL MOMENT IN TOP FACE)
Max BM (-) (M_x)=	137.576	kN m/m (LONGITUDINAL MOMENT IN BOTTOM FACE)

Max Shear Qx =	0.47	N/mm^2
Max Shear Qy =	0.46	N/mm^2

Effective depth of Raft in Vertical direction = D417 mm

Effective depth of Raft in Horizontal direction = D401 mm

Moment	Mux (kN-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)	Ast (mm^2)
My(+)	117.65	0.68	0.90	0.002	909.66	750.60	D16	221.03	300	175	1148.93
My(-)	132.60	0.76	1.02	0.002	1027.32	750.60	D16	195.71	300	175	1148.93
Mx(+)	99.36	0.62	0.82	0.002	797.83	721.80	D16	252.01	300	175	1148.93
Mx(-)	137.58	0.86	1.14	0.003	1110.87	721.80	D16	180.99	300	175	1148.93

Check against one-way shear force

Allowable shear stress, (ϕ Vc)/bd = 0.74 N/mm^2

Shear stress at a distance 'd' from raft top , τ_v= 0.47 N/mm^2 **OK**

SUMMARY OF REINFORCEMENT IN RAFT: **500 mm**

(HORIZONTAL REIN. IN TOP FACE)	**D16**	**175**	**c/c**
(HORIZONTAL REIN. IN BOTTOM FACE)	**D16**	**175**	**c/c**
(LONGITUDINAL REIN. IN TOP FACE)	**D16**	**175**	**c/c**
(LONGITUDINAL REIN. IN BOTTOM FACE)	**D16**	**175**	**c/c**

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Plate Center Stress Summary

			Shear		Membrane			Bending		
	Plate	L/C	Qx (N/mm^2)	Qy (N/mm^2)	Sx (N/mm^2)	Sy (N/mm^2)	Sxy (N/mm^2)	Mx (kNm/m)	My (kNm/m)	Mxy (kNm/m)
Max Qx	4508	206:1.4 (D + F	**0.026**	0.352	-0.341	-0.243	-0.052	24.004	134.769	2.163
Min Qx	4529	206:1.4 (D + F	**-0.026**	0.352	-0.340	-0.244	0.053	23.968	134.820	-2.158
Max Qy	4527	206:1.4 (D + F	0.005	**0.371**	-0.330	-0.319	-0.008	28.856	141.792	0.204
Min Qy	4714	204:1.2 (D + F	0.001	**-0.029**	0.720	-0.004	-0.005	-1.920	-2.304	-0.206
Max Sx	4718	206:1.4 (D + F	-0.001	-0.009	**1.207**	-0.161	0.001	4.754	2.989	0.050
Min Sx	4497	206:1.4 (D + F	0.003	0.325	**-0.349**	-0.137	-0.006	16.705	124.183	0.266
Max Sy	4505	205:0.9 (D + F	-0.004	0.192	-0.165	**0.042**	0.008	9.347	65.329	-0.275
Min Sy	4527	204:1.2 (D + F	0.005	0.301	-0.293	**-0.419**	-0.040	25.456	121.339	-0.148
Max Sxy	4495	204:1.2 (D + F	-0.020	0.268	-0.261	-0.133	**0.101**	16.394	100.338	-1.224
Min Sxy	4516	204:1.2 (D + F	0.020	0.268	-0.261	-0.133	**-0.100**	16.405	100.342	1.234
Max Mx	4501	206:1.4 (D + F	0.005	0.371	-0.331	-0.318	-0.008	**28.865**	141.780	0.202
Min Mx	4653	206:1.4 (D + F	0.001	0.007	0.710	-0.161	-0.003	**-4.381**	0.338	-0.096
Max My	4527	206:1.4 (D + F	0.005	0.371	-0.330	-0.319	-0.008	28.856	**141.792**	0.204
Min My	4662	205:0.9 (D + F	0.001	-0.001	0.335	0.028	-0.007	-2.495	**-8.326**	-0.268
Max Mxy	4650	204:1.2 (D + F	-0.009	0.008	0.568	-0.133	0.079	2.037	-1.150	**2.932**
Min Mxy	4673	204:1.2 (D + F	0.009	0.008	0.568	-0.133	-0.079	2.055	-1.146	**-2.927**

Plate Center Stress Summary

			Shear		Membrane			Bending		
	Plate	L/C	Qx (N/mm^2)	Qy (N/mm^2)	Sx (N/mm^2)	Sy (N/mm^2)	Sxy (N/mm^2)	Mx (kNm/m)	My (kNm/m)	Mxy (kNm/m)
Max Qx	5248	206:1.4 (D + F	**0.466**	0.017	0.000	0.000	0.000	133.540	36.971	7.558
Min Qx	5469	206:1.4 (D + F	**-0.466**	-0.046	0.000	0.000	0.000	132.929	38.301	13.418
Max Qy	5411	206:1.4 (D + F	0.044	**0.456**	0.000	0.000	0.000	50.051	130.048	30.280
Min Qy	5356	206:1.4 (D + F	-0.036	**-0.455**	0.000	0.000	0.000	36.914	132.597	-7.702
Max Sx	6211	201:1.4 (D + F	0.017	0.261	**0.000**	0.000	0.000	0.057	-98.876	-5.765
Min Sx	6211	201:1.4 (D + F	0.017	0.261	**0.000**	0.000	0.000	0.057	-98.876	-5.765
Max Sy	6211	201:1.4 (D + F	0.017	0.261	0.000	**0.000**	0.000	0.057	-98.876	-5.765
Min Sy	6211	201:1.4 (D + F	0.017	0.261	0.000	**0.000**	0.000	0.057	-98.876	-5.765
Max Sxy	6211	201:1.4 (D + F	0.017	0.261	0.000	0.000	**0.000**	0.057	-98.876	-5.765
Min Sxy	6211	201:1.4 (D + F	0.017	0.261	0.000	0.000	**0.000**	0.057	-98.876	-5.765
Max Mx	5246	206:1.4 (D + F	0.455	-0.043	0.000	0.000	0.000	**137.576**	41.239	13.318
Min Mx	5197	206:1.4 (D + F	-0.271	-0.012	0.000	0.000	0.000	**-99.363**	1.333	-0.373
Max My	5356	206:1.4 (D + F	-0.036	-0.455	0.000	0.000	0.000	36.914	**132.597**	-7.702
Min My	5183	206:1.4 (D + F	-0.002	0.305	0.000	0.000	0.000	-0.423	**-117.649**	2.742
Max Mxy	6101	206:1.4 (D + F	-0.154	0.192	0.000	0.000	0.000	-40.297	-36.523	**53.567**
Min Mxy	6066	206:1.4 (D + F	0.154	0.192	0.000	0.000	0.000	-40.296	-36.521	**-53.567**

REFERENCES

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

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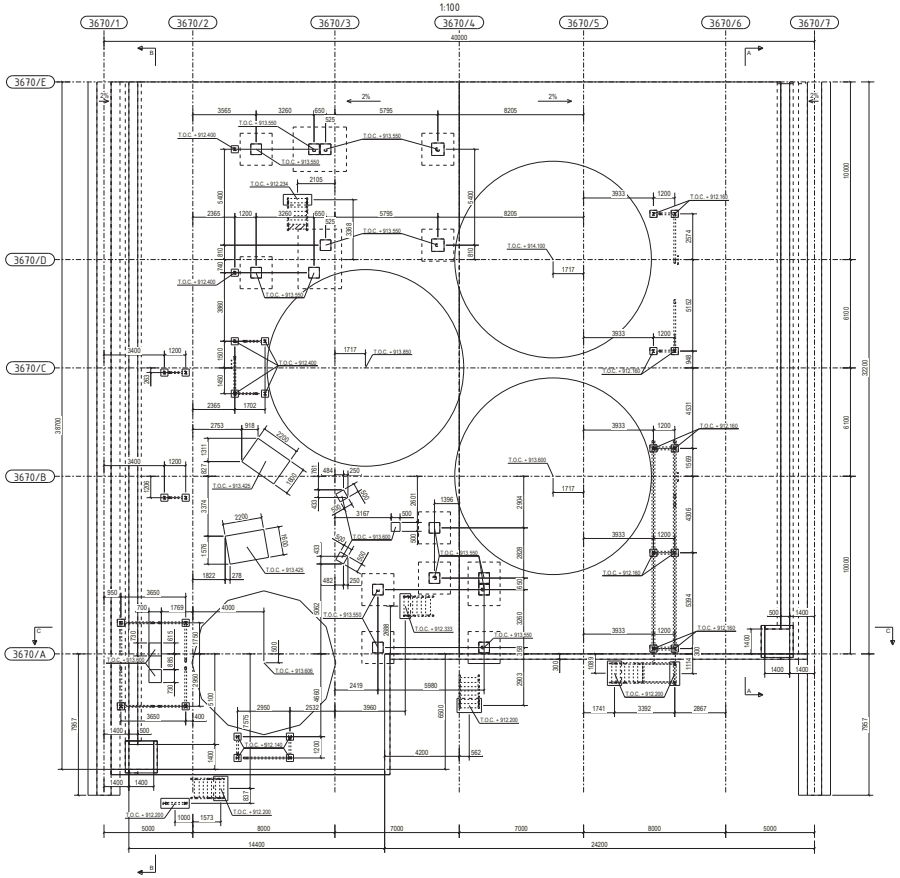
ATTACHMENT 1 LOAD DATA DRAWINGS

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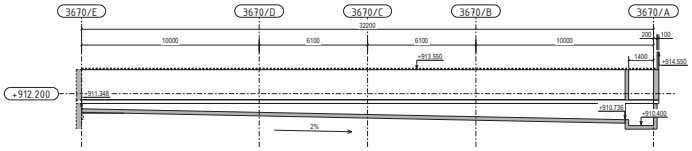
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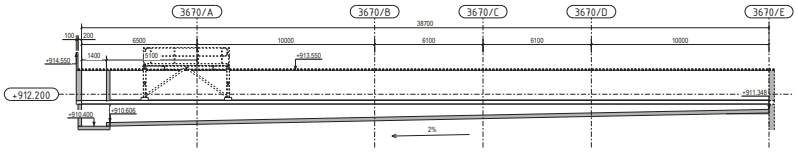
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A – A
1:100



B – B
1:100



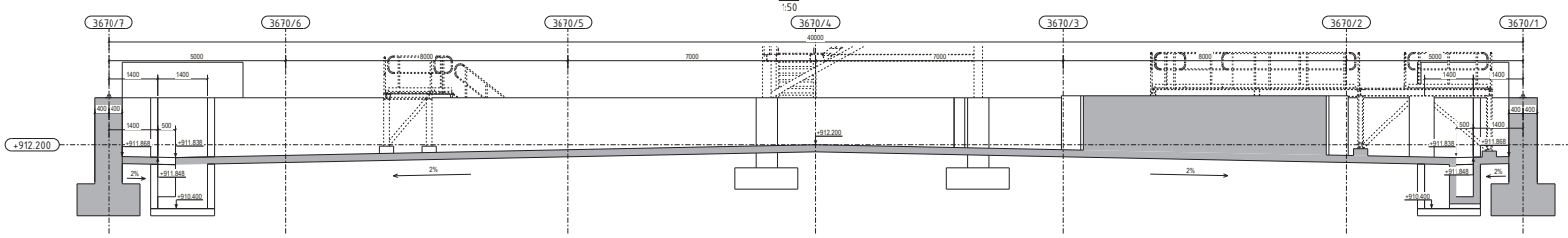
THIS DESIGN IS NOT ISSUED FOR CONSTRUCTION

FINISHED TOP GROUND LEVEL (F.G.L) FOR PROCESS AREA 3670 = RL(+)912.200 m

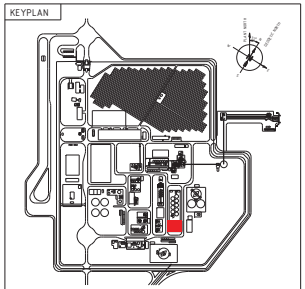
CONCRETE ELEMENT FLOOR (PRE-CAST SLAB)
CONCRETE FLOOR
CONCRETE FLOOR WITH ACID RESISTANT COATING
CONCRETE WALL

1. ALL DIMENSIONS ARE IN MILLIMETER, ELEVATIONS ARE IN METER UNLESS OTHERWISE STATED
2. CO-ORDINATES ARE SHOWN IN GENERAL ARRENGEMENT DRAWINGS
3. GRATING TYPE (FRP OR NORMAL) IS SPECIFIED IN GENERAL ARRENGEMENT DRAWINGS
4. F.F.L, F.G.L, T.O.C = FINISHED FLOOR LEVEL, FINISHED GROUND LEVEL, TOP OF CONCRETE, T.O.S, B.O.S, B.O.B, = TOP OF STEEL, BOTTOM OF STEEL, BOTTOM OF BASE PLATE
5. BUILDING DIMENSIONS AND HEIGHTS ARE TENTATIVE, SHALL BE FINALIZED DURING DETAIL ENGINEERING SUBJECT TO MINIMUM AS SHOWN.
6. ALL MAINTENANCE FACILITIES, HANDLING PROVISIONS, ACCESS WAYS, PLATFORMS FOR MAINTENANCE SHALL BE PROVIDED DURING DETAIL ENGINEERING WHICH ARE NOT SHOWN ENTIRELY. THIS DRAWING IS TO BE CONSIDERED AS CONCEPTUAL ARRANGEMENT OF THE FACILITIES.

C – C
1:50



KEYPLAN

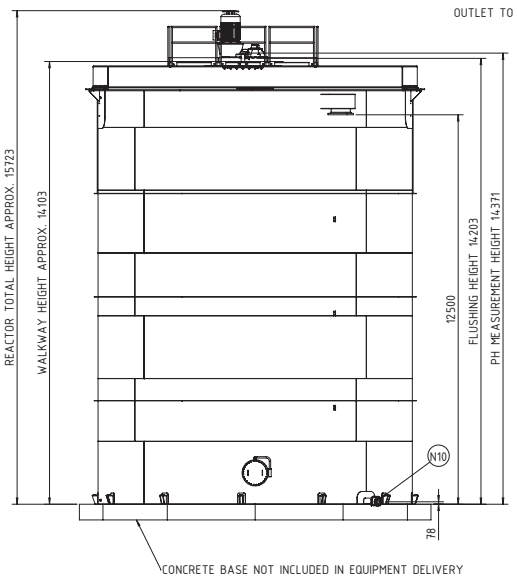


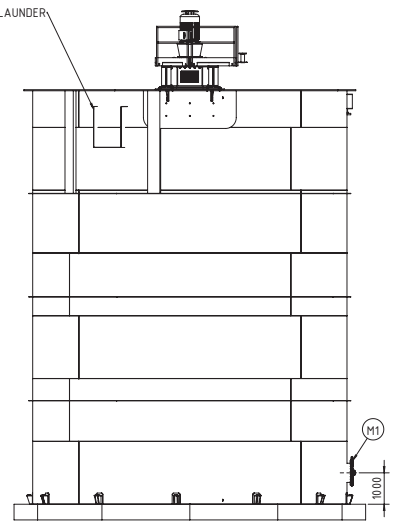
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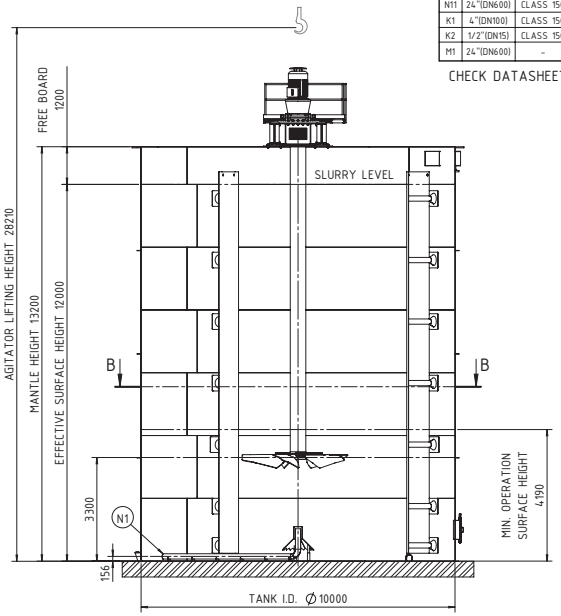
DRAWING NUMBER MD-216-3670-EG-ME-GAD-1002 REV A01

NOZZLE TABLE								
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N1	6"(DN150)	CLASS 150	300°	156	ASME B16.5	AIR INLET	1	
N10	6"(DN150)	CLASS 150	45°	78	ASME B16.5	DRAIN	1	
N11	24"(DN600)	CLASS 150	60°	12500	ASME B16.5	OVERFLOW	1	
K1	4"(DN100)	CLASS 150	90°	14371	ASME B16.5	PH MEASUREMENT	1	
K2	1/2"(DN15)	CLASS 150	0°	14203	ASME B16.5	FLUSHING	1	
M1	24"(DN600)	-	90°	1000	OUTOTEC	MANHOLE	1	

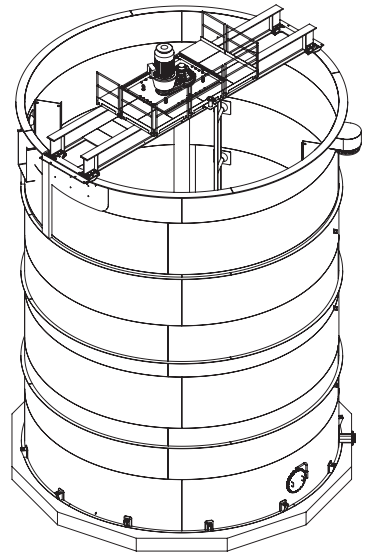
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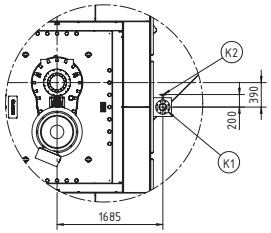




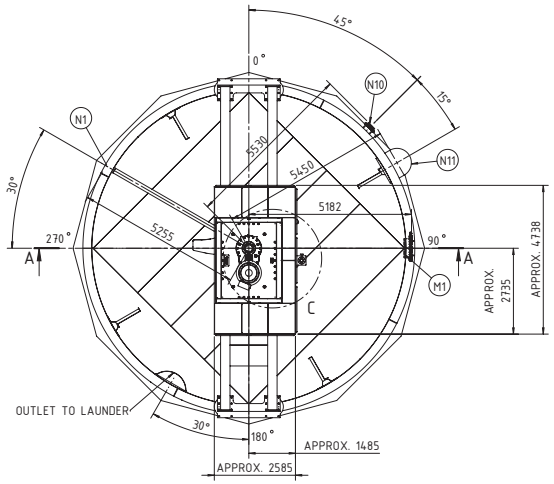


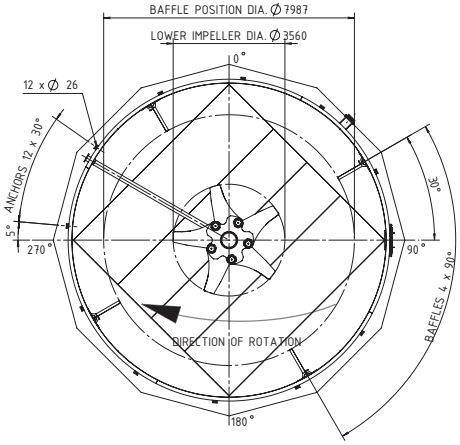
SECTION A-A





DETAIL C
SCALE 1 : 40





SECTION B-B

CYANIDE DESTRUCTION REACTOR	
DEAD WEIGHT (kg)	32570
OPERATIONAL WEIGHT (kg)	1380154
MAXIMUM WEIGHT (kg)	1513164

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3670-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3670-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3670-EG-PR-PID-1001	P&I DIAGRAM								
MD-216-3670-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	14.01.2020	ISSUED FOR APPROVAL	AT	TM	JR	GR	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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APPROVED	GeoRie	14.01.2020
EMPLOYER		
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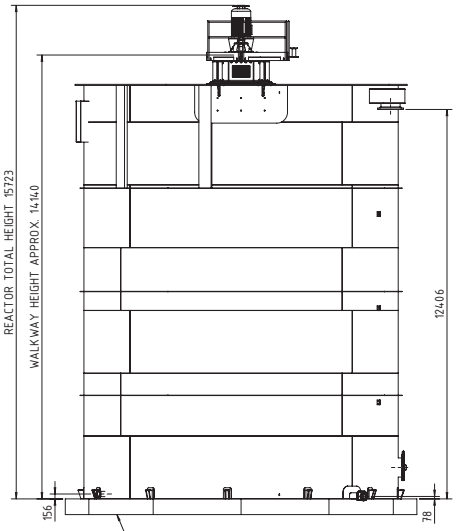
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PRIMARY CYANIDE DESTRUCTION
3670-REA-001 CYANIDE DESTRUCTION REACTOR

DRAWING NUMBER MD-216-3670-EG-ME-GAD-1002 REV A01

DRAWING NUMBER MD-216-3670-EG-ME-GAD-1003 REV A01

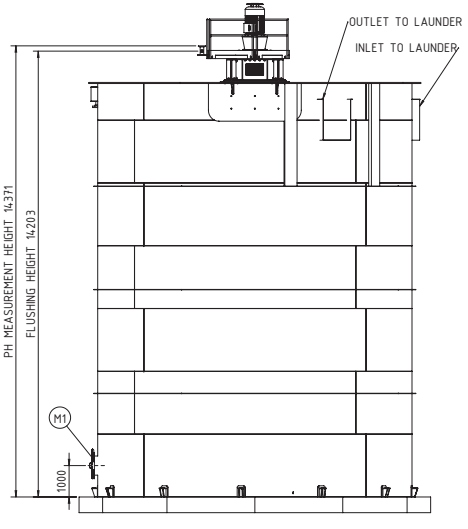
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	6"(DN150)	CLASS 150	60°	156	ASME B16.5	AIR INLET	1	
N10	6"(DN150)	CLASS 150	315°	78	ASME B16.5	DRAIN	1	
N11	24"(DN600)	CLASS 150	300°	12406	ASME B16.5	OVERFLOW	1	
K1	4"(DN100)	CLASS 150	270°	14371	ASME B16.5	PH MEASUREMENT	1	
K2	1/2"(DN15)	CLASS 150	180°	14203	ASME B16.5	FLUSHING	1	
M1	24"(DN600)	–	270°	1000	OUTOTEC	MANHOLE	1	

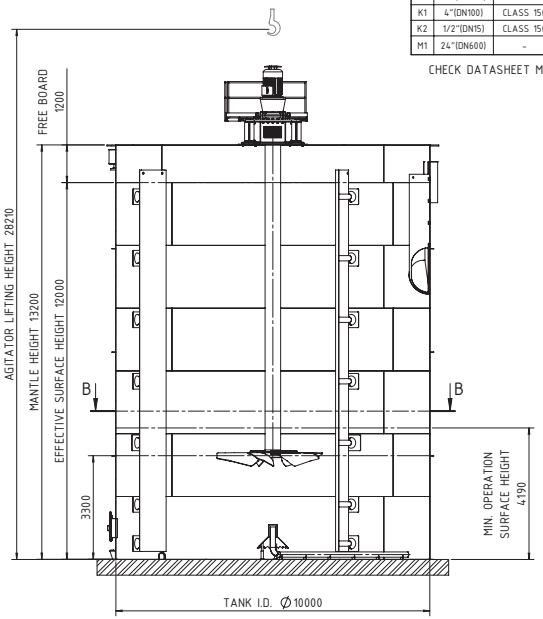
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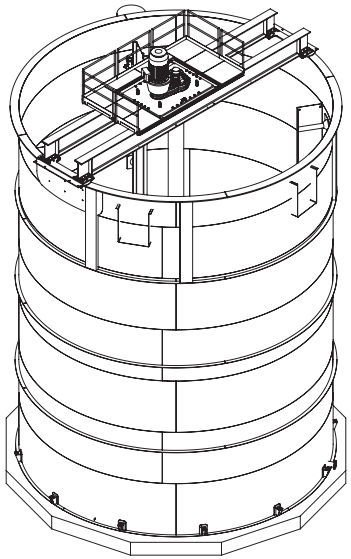
SECTION C-C

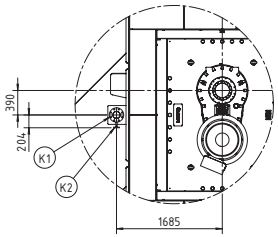
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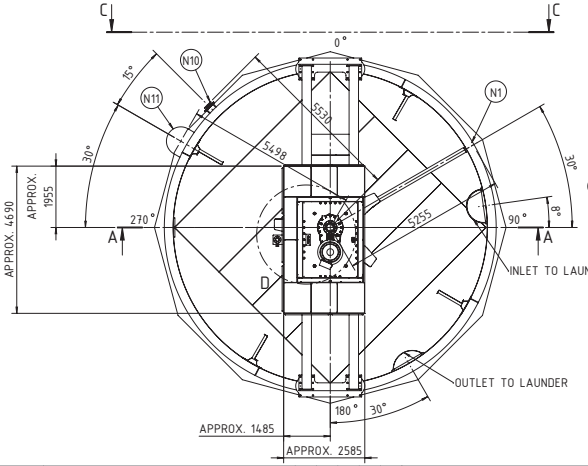


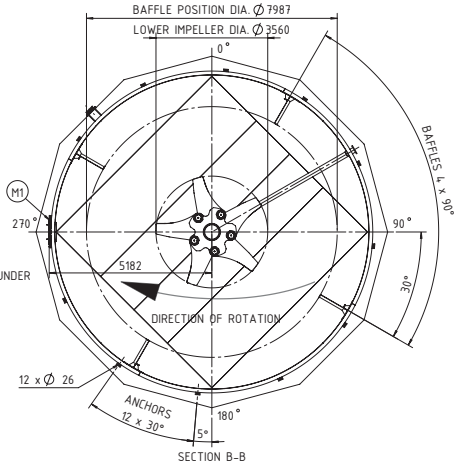
SECTION A-A





DETAIL D
SCALE 1 : 40





SECTION B-B

CYANIDE DESTRUCTION REACTOR	
DEAD WEIGHT (kg)	32570
OPERATIONAL WEIGHT (kg)	1380154
MAXIMUM WEIGHT (kg)	1513164

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3670-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3670-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3670-EG-PR-PID-1002	P&I DIAGRAM								
MD-216-3670-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	14.01.2020	ISSUED FOR APPROVAL	AT	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02923-002 REVISION 0
OUTOTEC DRAWING NUMBER: 900707 GAU01 RN12 MTL03 36702 DOCUMENT ID OU601246276 REVISION 0

DRAWN	aritik	09.01.2020
CHECKED	tuoman	14.01.2020
ENGINEER	juhris	14.01.2020
APPROVED	GeoRie	14.01.2020
EMPLOYER		
SCALE	1:80	A1

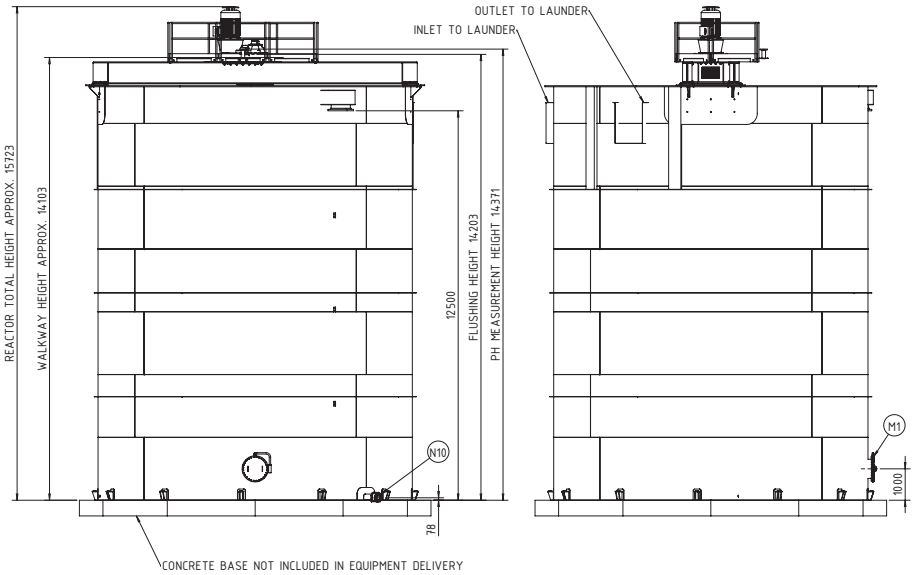
MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
PRIMARY CYANIDE DESTRUCTION
3670-REA-002 CYANIDE DESTRUCTION REACTOR

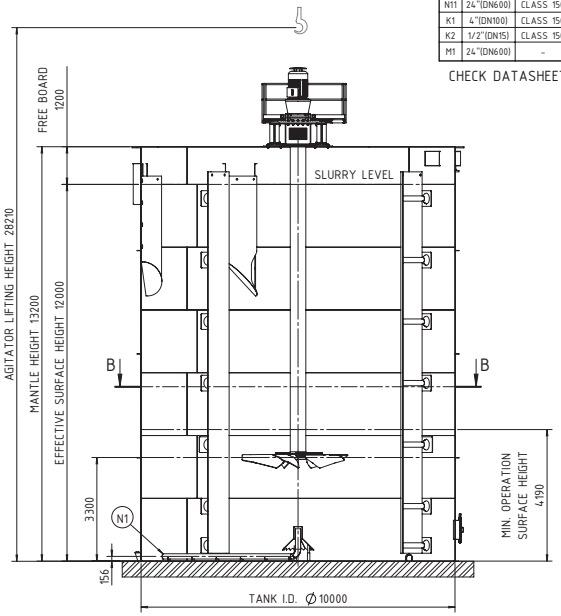
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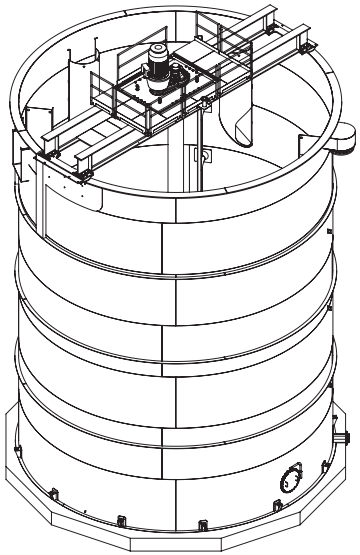
NOZZLE TABLE								
No.	SIZE	RATING	DIRECTION	HEIGHT	STANDARD	SERVICE	QUANTITY	REMARKS
N1	6"(DN150)	CLASS 150	300°	156	ASME B16.5	AIR INLET	1	
N10	6"(DN150)	CLASS 150	45°	78	ASME B16.5	DRAIN	1	
N11	24"(DN600)	CLASS 150	60°	12500	ASME B16.5	OVERFLOW	1	
K1	4"(DN100)	CLASS 150	90°	14371	ASME B16.5	PH MEASUREMENT	1	
K2	1/2"(DN15)	CLASS 150	0°	14203	ASME B16.5	FLUSHING	1	
M1	24"(DN600)	-	90°	1000	OUTOTEC	MANHOLE	1	

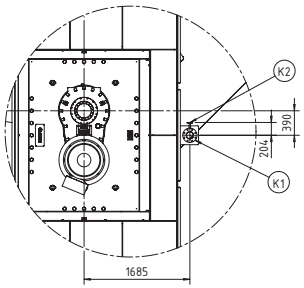
CHECK DATASHEET MD-216-3670-EG-ME-DAT-1001 FOR PROCESS VALUES



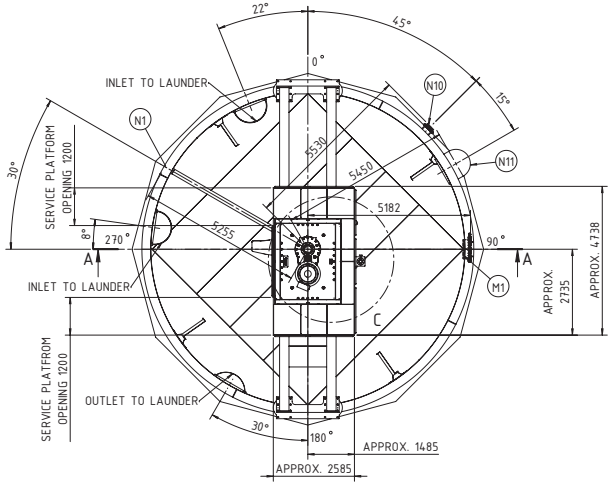


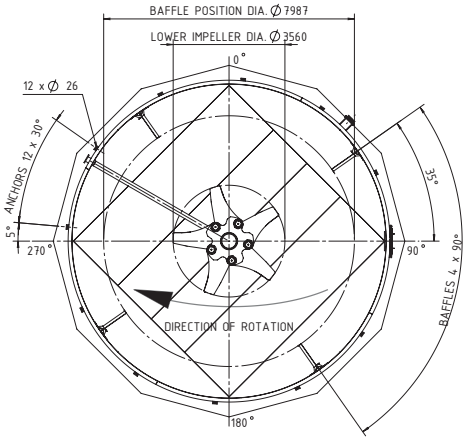
SECTION A-A





DETAIL C
SCALE 1 : 40





SECTION B-B

CYANIDE DESTRUCTION REACTOR	
DEAD WEIGHT (kg)	32570
OPERATIONAL WEIGHT (kg)	1380154
MAXIMUM WEIGHT (kg)	1513164

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3670-EG-ME-PLN-1001	FOUNDATION AND LOAD PLAN								
MD-216-3670-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3670-EG-PR-PID-1003	P&I DIAGRAM								
MD-216-3670-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	14.01.2020	ISSUED FOR APPROVAL	AT	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-G-02923-003 REVISION 0

OUTOTEC DRAWING NUMBER: 900707 GAU01 RN12 MTL03 36703 DOCUMENT ID OU601388348 REVISION 0

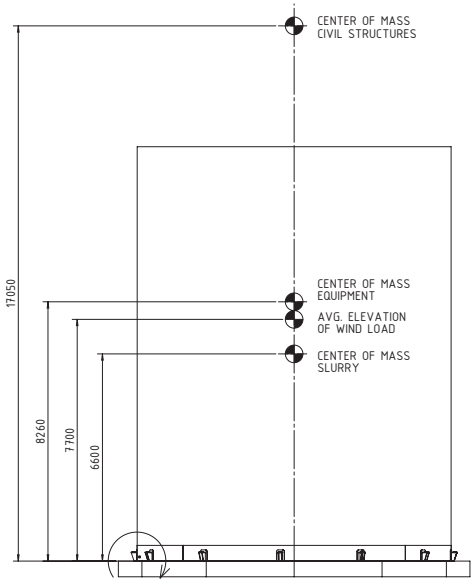
DRAWN	aritik	09.01.2020
CHECKED	tuoman	14.01.2020
ENGINEER	juhris	14.01.2020
APPROVED	GeoRie	14.01.2020
EMPLOYER		
SCALE	1:80	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
CYANIDE DETOXIFICATION
3670-REA-003 CYANIDE DESTRUCTION REACTOR

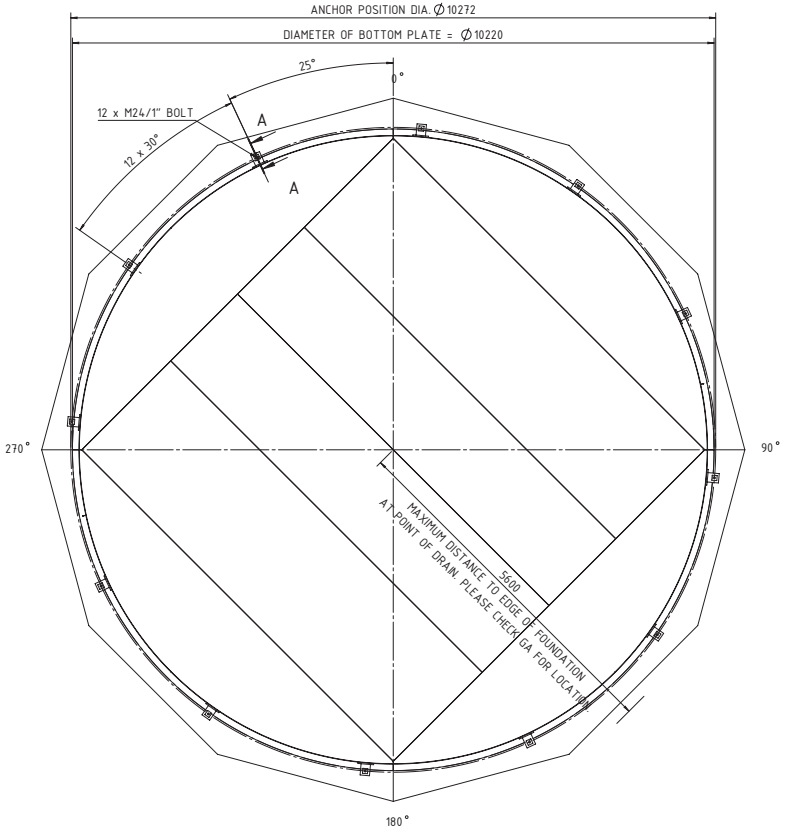
DRAWING NUMBER MD-216-3670-EG-ME-GAD-1004 REV A01

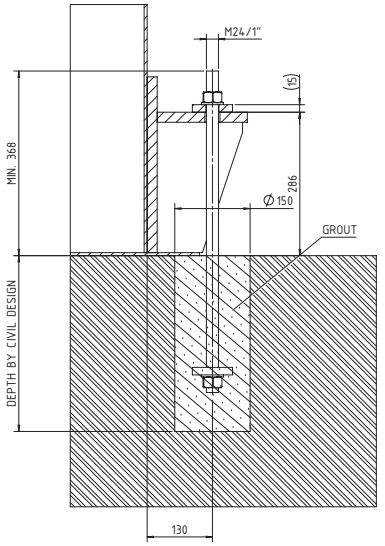
DRAWING NUMBER MD-216-3670-EG-ME-PLN-1001 REV A01

LOADING NOTES:	
1	THE LOWEST NATURAL FREQUENCY OF COMBINED MACHINE, FLUID LOAD, AND SUPPORT SYSTEM IN ANY DIRECTION MUST EXCEED 1.5 TIMES THE ROTOR ROTATIONAL FREQUENCY (REFER NORMAL OPERATING SPEED*)
2	ALL LOADS ASSUME UNIFORM SUPPORT OF TANK BASE. (TANK LOCATED ON THE GROUND OR CONCRETE SLAB)
FOUNDATION AND SUPPORT TOLERANCES	
1.	THE FIRST 300mm WIDTH OF AN ANNULAR RING MEASURED FROM THE OUTSIDE OF THE TANK RADIALLY TOWARDS THE CENTRE SHALL BE WITHIN +/-3mm OF THE DESIGN LEVEL
2.	THE REMAINDER OF THE FOUNDATION SHALL BE WITHIN +/-6mm OF THE DESIGN LEVEL
CONCRETE NOTES: (WHERE CELLS ARE INSTALLED ON CONCRETE AND HOLD DOWN BOLTS ARE USED)	
1.	CONCRETE FOUNDATION OUTLINE SHOWN GIVES NOMINAL EDGE DISTANCE FOR HOLD-DOWN BOLT TO EDGE OF FOUNDATION AT 110mm
2.	FINAL OVERALL CONCRETE FOUNDATION DIMENSION AND REQUIRED HOLD-DOWNBOLT EDGE DISTANCE TO BE DETERMINED BY CLIENT



SCALE 1 : 80





SECTION A-A
SCALE 1 : 5

CYANIDE DESTRUCTION TANK	
DEAD WEIGHT OF EQUIPMENT	64000 kg
DEAD WEIGHT OF CIVIL STRUCTURES	125000 kg
OPERATIONAL WEIGHT OF SLURRY	1316100 kg
MAXIMUM WEIGHT OF SLURRY	1449100 kg
EXTERNAL PLATFORM LOAD	10000 kg
NORMAL OPERATING SPEED	32 rpm
WIND LOAD	154,7 kN
SEISMIC LOAD	643 kN
SEISMIC TORQUE AT FOUNDATION LEVEL	5988 kNm

SEISMIC LOAD IS CALCULATED WITH FOLLOWING VALUES
-S1=0,02g
-Ss=0,10g
-max. groud elevation 3m
CIVIL ENGINEERING TO CHECK SEISMIC LOAD AT ACTUAL ELEVATION

WIND LOAD IS CALCULATED WITH WIND SPEED OF 43 m/s

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-3670-EG-ME-GAD-1002...1004	GENERAL ARRANGEMENT								
MD-216-3670-EG-ME-DAT-1001	EQUIPMENT DATASHEET								
MD-216-3670-EG-PR-PID-1001...1003	P&I DIAGRAM								
MD-216-3670-EG-PR-PFD-1001	PROCESS FLOW DIAGRAM	A01	14.01.2020	ISSUED FOR APPROVAL	JJ	TM	JR	GR	

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LARSEN & TOUBRO DRAWING NUMBER: 019045-450-M-D-02924-001 REVISION: 1

OUTOTEC DRAWING NUMBER: 900707 GAU01 RN12 MLD15 36701 DOCUMENT ID: OU601268812 REVISION: 1

DRAWN	joujaa	10.01.2020
CHECKED	tuoman	14.01.2020
ENGINEER	juhris	14.01.2020
APPROVED	GeoRie	14.01.2020
EMPLOYER		
SCALE	1:40	A1

MANSOURAH - MASSARAH GOLD PROJECT
FOUNDATION AND LOAD PLAN
PRIMARY CYANIDE DESTRUCTION
3670-REA-001...003 CYANIDE DESTRUCTION REACTOR

DRAWING NUMBER: MD-216-3670-EG-ME-PLN-1001 REV: A01

ATTACHMENT 2 STAAD OUTPUT

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```
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*           Version  20.07.10.64                 *
*           Proprietary Program of               *
*           Bentley Systems, Inc.                *
*           Date=    FEB 26, 2020                *
*           Time=    13:19:50                    *
*                                                *
*      USER ID: LNT ECC DIV                      *
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     3. ENGINEER DATE 25.02.2020
     4. JOB NAME MANSOURAH MASSARAH GOLD PROJECT
     5. JOB CLIENT MA'ADEN
     6. ENGINEER NAME SOR
     7. CHECKER NAME SJM
     8. APPROVED NAME AG
     9. JOB REV 0
    10. END JOB INFORMATION
    11. INPUT WIDTH 79
    12. UNIT MMS KN
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542. 6211 THICKNESS 500
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548. ALPHA 1E-005
549. DAMP 0.05
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551. STRENGTH FCU 0.027579
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561. 5959 TO 6211
562. JOINT
563. END GROUP DEFINITION
564. UNIT METER KN
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838. 4752 TRAP JT 0 0 -6 -6
839. 4753 TRAP JT 0 0 -6 -6
840. ********************************************************************************
841. LOAD 2 LOADTYPE FLUIDS  TITLE OPERATIONAL FLUID LOAD ( F1 )
842. ELEMENT LOAD
843. 4494 TO 4753 PR -83.83
844. 5049 TO 5056 5161 TO 5169 5171 5173 5176 5178 5180 5182 TO 5185 5187 5189 -
845. 5190 5193 TO 5195 5197 5199 5200 5202 5958 TO 6211 PR GY -167.65
846. ********************************************************************************
847. LOAD 3 LOADTYPE FLUIDS  TITLE MAX FLUID LOAD ( F2 )
848. ELEMENT LOAD
849. 4494 TO 4753 PR -92.3
850. 5049 TO 5056 5161 TO 5169 5171 5173 5176 5178 5180 5182 TO 5185 5187 5189 -
851. 5190 5193 TO 5195 5197 5199 5200 5202 5958 TO 6211 PR GY -184.6
852. ********************************************************************************
853. LOAD 4 LOADTYPE WIND  TITLE WIND LOAD ( W )
854. JOINT LOAD
855. 2462 TO 2513 FZ -2.975
856. 2487 FY 0
857. 2488 FY -1.1
858. 2489 FY -2.2
859. 2490 FY -3.2
860. 2491 FY -4.3
861. 2492 FY -5.2
862. 2493 FY -6.1
863. 2494 FY -6.9
864. 2495 FY -7.5
865. 2496 FY -8.1
866. 2497 FY -8.6
867. 2498 FY -8.9
868. 2499 FY -9.1
869. 2500 FY -9.2
870. 2501 FY -9.1
871. 2502 FY -8.9
872. 2503 FY -8.6
873. 2504 FY -8.1
874. 2505 FY -7.5
875. 2506 FY -6.9
876. 2507 FY -6.1
877. 2508 FY -5.2
878. 2509 FY -4.3
```

879. 2510 FY -3.2
880. 2511 FY -2.2
881. 2512 FY -1.1
882. 2513 FY 0
883. 2462 FY 1.1
884. 2463 FY 2.2
885. 2464 FY 3.2
886. 2465 FY 4.3
887. 2466 FY 5.2
888. 2467 FY 6.1
889. 2468 FY 6.9
890. 2469 FY 7.5
891. 2470 FY 8.1
892. 2471 FY 8.6
893. 2472 FY 8.9
894. 2473 FY 9.1
895. 2474 FY 9.2
896. 2475 FY 9.1
897. 2476 FY 8.9
898. 2477 FY 8.6
899. 2478 FY 8.1
900. 2479 FY 7.5
901. 2480 FY 6.9
902. 2481 FY 6.1
903. 2482 FY 5.2
904. 2483 FY 4.3
905. 2484 FY 3.2
906. 2485 FY 2.2
907. 2486 FY 1.1
908. **********
909. LOAD 5 LOADTYPE SEISMIC TITLE SEISMIC LOAD (E)
910. JOINT LOAD
911. 2462 TO 2513 FZ -12.365
912. 2487 FY 0
913. 2488 FY -5.6
914. 2489 FY -11
915. 2490 FY -16.3
916. 2491 FY -21.4
917. 2492 FY -26.2
918. 2493 FY -30.5
919. 2494 FY -34.5
920. 2495 FY -37.9
921. 2496 FY -40.8
922. 2497 FY -43.1
923. 2498 FY -44.7
924. 2499 FY -45.7
925. 2500 FY -46.1
926. 2501 FY -45.7
927. 2502 FY -44.7
928. 2503 FY -43.1
929. 2504 FY -40.8
930. 2505 FY -37.9
931. 2506 FY -34.5
932. 2507 FY -30.5
933. 2508 FY -26.2
934. 2509 FY -21.4

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935. 2510 FY -16.3
936. 2511 FY -11
937. 2512 FY -5.6
938. 2513 FY 0
939. 2462 FY 5.6
940. 2463 FY 11
941. 2464 FY 16.3
942. 2465 FY 21.4
943. 2466 FY 26.2
944. 2467 FY 30.5
945. 2468 FY 34.5
946. 2469 FY 37.9
947. 2470 FY 40.8
948. 2471 FY 43.1
949. 2472 FY 44.7
950. 2473 FY 45.7
951. 2474 FY 46.1
952. 2475 FY 45.7
953. 2476 FY 44.7
954. 2477 FY 43.1
955. 2478 FY 40.8
956. 2479 FY 37.9
957. 2480 FY 34.5
958. 2481 FY 30.5
959. 2482 FY 26.2
960. 2483 FY 21.4
961. 2484 FY 16.3
962. 2485 FY 11
963. 2486 FY 5.6
964. ********************************************************************************
965. * LOAD COMBINATIONS
966. LOAD COMB 201 1.4 ( D + F1 )
967. 1 1.4 2 1.4
968. LOAD COMB 202 0.9 D + 1.6 W
969. 1 0.9 4 1.6
970. LOAD COMB 203 1.2 ( D + F1 ) + 1.6 W
971. 1 1.2 2 1.2 4 1.6
972. LOAD COMB 204 1.2 ( D + F1 ) + E
973. 1 1.2 2 1.2 5 1.0
974. LOAD COMB 205 0.9 ( D + F1 ) + E
975. 1 0.9 2 0.9 5 1.0
976. LOAD COMB 206 1.4 ( D + F2 )
977. 1 1.4 3 1.4
978. PERFORM ANALYSIS
```

P R O B L E M S T A T I S T I C S

NUMBER OF JOINTS	**712**	**NUMBER OF MEMBERS**	**0**
NUMBER OF PLATES	**706**	**NUMBER OF SOLIDS**	**0**
NUMBER OF SURFACES	**0**	**NUMBER OF SUPPORTS**	**400**

SOLVER USED IS THE OUT-OF-CORE BASIC SOLVER

ORIGINAL/FINAL BAND-WIDTH= 692/ 41/ 180 DOF
TOTAL PRIMARY LOAD CASES = 5, TOTAL DEGREES OF FREEDOM = 3072
TOTAL LOAD COMBINATION CASES = 6 SO FAR.
SIZE OF STIFFNESS MATRIX = 553 DOUBLE KILO-WORDS
REQRD/AVAIL. DISK SPACE = 21.8/ 187098.3 MB

979. FINISH

********** END OF THE STAAD.Pro RUN **********

**** DATE= FEB 26,2020 TIME= 13:20: 7 ****

****
* For technical assistance on STAAD.Pro, please visit *
* http://selectservices.bentley.com/en-US/ *
* *
* Details about additional assistance from *
* Bentley and Partners can be found at program menu *
* Help->Technical Support *
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